One Shot, Twenty-One Balls: Existence and Rarity of a Total Clearance in a Single Stroke of Snooker

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Abstract

Snooker folklore holds that no single stroke can pocket all twenty-one object balls. We examine the claim in an idealized but fully specified model of billiard dynamics. Within the model we exhibit an admissible configuration of the twenty-two balls and a stroke of the cue ball that pockets all twenty-one object balls, and we show that the set of such strokes has positive Lebesgue measure in the natural shot space: total clearances are not flukes of measure zero but open events. For the regulation opening configuration we conjecture the same and explain both why a simulation cannot settle the conjecture by brute force and what kind of computation could settle it in principle. Monte Carlo experiments in the same model estimate the probability $P(k)$ that a uniformly random stroke pockets exactly k balls; the observed decay of $P(k)$, extrapolated conditionally on the conjecture, places the probability of a total clearance from the break far beyond anything observable. The folk claim is thus right in practice and wrong in principle, and the gap between the two is exactly the distance between measure zero and unobservably small.

1 Introduction

Ask a snooker player whether one stroke can sink all twenty-one object balls and you will get a smile and a firm no. The smile is justified; we will argue the no is not, at least not in the sense usually meant. There are three different claims hiding in the folklore: (i) no legal stroke can possibly pot all twenty-one balls; (ii) from the regulation opening position, no stroke pots all twenty-one balls; (iii) you will never see it happen. Claim (iii) is true beyond reasonable doubt, and we will quantify just how true. Claim (i) is false in any reasonable idealization of the game, and the point of this article is that it fails in a robust way *in principle*: the successful strokes form a set of positive measure, so a total clearance is not a lone knife-edge coincidence but occupies genuine volume in the shot space. That volume, we will see, can be astronomically small—for a twenty-one-ball clearance it is far below any humanly or even numerically resolvable width—so "positive measure" and "observably robust" part company, and the gap between them is one of the things this article is about. Claim (ii) we believe to be false as well, but the question turns out to sit in an interesting computational blind spot: it is decidable in principle by a finite certified computation, yet hopeless to decide by direct sampling.

The mathematical content is elementary in the best sense. Billiard dynamics between collisions is smooth and explicit; a total clearance designed with strict margins survives small perturbations because a finite composition of continuous maps is continuous; and an open nonempty set has positive measure. The pleasure is in the construction, in saying precisely what model makes the statement true, and in seeing how the same model measures the astronomical rarity that protects the folklore. Mathematically rigorous statements about idealized billiards have a long tradition [1, 2]; our contribution is to point that tradition at a bar bet.

2 The model

We work with a fully explicit model, called Model M below. Stating it precisely is not pedantry: we will see in Remark 2 that the standard opening position of snooker, taken literally, does not even define a dynamical system, so some care about the model is forced on us.

Definition 1 (Table, balls, pockets)**.** The table is the closed rectangle $T = [0, L] \times [0, W]$ with $L = 3.569$ and $W = 1.778$ (meters; the playing area of a full-size table). Balls are closed disks of common radius $R = 0.02625$. There are six pocket points on the boundary: the four corners and the two midpoints of the long sides. A ball is *pocketed* (and removed from the table) at the first instant its center enters the open disk of radius $\rho = 0.055$ about a pocket point.

Definition 2 (Dynamics)**.** Between events, a moving ball decelerates at the constant rate $a = 0.10\ \mathrm{m/s}^2$ antiparallel to its velocity, so its center travels along a straight segment, quadratically in time, until it stops. Ball-ball collisions are instantaneous, frictionless, and equal-mass, with normal restitution $e = 0.95$: the velocity components along the line of centers exchange in the usual restitution combination, tangential components are unchanged. A cushion reverses the normal velocity component and scales it by $e_c = 0.90$. Spin is absent from the model; Remark 1 discusses why this loses nothing for our purposes.

Definition 3 (Configurations and shots)**.** A *configuration* c places the cue ball and the twenty-one object balls at pairwise distances strictly greater than $2R$, all disks inside T, no center inside a pocket disk. We call such c *admissible*. A *shot* is a pair $x = (V_0, \varphi) \in \Omega = (0, V_{\max}] \times [0, 2\pi)$ with $V_{\max} = 12$ m/s: the cue ball is set in motion with speed V_0 in direction φ. Given an admissible c and a shot x, the dynamics of Definition 2 runs until all balls are at rest or pocketed; $N(c, x) \in \{0, 1, \dots, 21\}$ denotes the number of object balls pocketed.

We chose to say *admissible* rather than *legal*. A legal position of snooker is one reachable from the opening position by legal play, and we make no claim that our witness configuration is reachable in that sense; nothing in the mathematics needs it, and the folklore claim (i) concerns physical possibility, not the rule book.

Remark 1 (Why no spin)*.* Real strokes impart follow, draw, and side, and real balls slide before they roll. Including spin would enlarge the shot space by two or more dimensions and complicate the collision law, but it would not change the logic of anything below: the existence theorem needs only that the final state depends continuously on the shot near one well-chosen shot, and that argument is indifferent to the dimension of the shot space or the detailed smooth collision law. A witness in the spinless model, struck with zero spin, remains a witness in any enlarged model whose dynamics agrees with Model M on the zero-spin slice. What genuinely changes with spin is the quantitative part, the estimated $P(k)$; we return to this in Section 5.

Remark 2 (The touching rack is not a dynamical system)*.* In the regulation opening position the fifteen reds touch. The first impact then produces simultaneous multi-ball contacts, and the outcome of a simultaneous contact is not determined by pairwise restitution laws: resolving the contacts in different micro-orders yields genuinely different outgoing velocities, a phenomenon well known in nonsmooth mechanics [3]. In other words, for the literal opening rack, $N(c_0, x)$ is not defined by the model until an arbitrary tie-breaking convention is chosen. We therefore work throughout with the ε-separated opening configuration c_0^ε, the regulation positions with the reds spread by $\varepsilon = 0.5$ mm so that all pairwise gaps are positive. This is not a dodge but a modeling necessity, and every statement about the break below refers to c_0^ε.

3 A total clearance exists, robustly

Call $x \in \Omega$ *regular* for the admissible configuration c if the resulting evolution consists of finitely many events (ball-ball collisions, cushion contacts, pocketings, stops), all *transversal*: at each ball-ball or cushion contact the normal approach speed is strictly positive, each pocketing crosses the pocket-disk boundary with strictly positive speed, no two events are simultaneous, and no ball comes to rest exactly at a contact.

Lemma 1 (Continuity off the singular set)**.** *Let c be admissible and let x^* be regular for c. Then $N(c, \cdot)$ is constant on a neighborhood of x^* in Ω.*

Proof. Between events the state (positions and velocities of the active balls) evolves by an explicit smooth flow: each moving center follows $p+vt-\frac{a}{2}t^2\hat{v}$ until its stopping time $|v|/a$. Each upcoming event time is the first zero of a smooth clock function of the state (pairwise gap minus $2R$; distance to a cushion line; distance to a pocket point minus ρ; a speed). Transversality says precisely that the clock has a simple zero with nonvanishing time derivative there, and that no other clock vanishes at the same instant. By the implicit function theorem the event time, and hence the state just before the event, depends continuously on the initial data; the event maps themselves (the restitution exchange, the cushion reflection, removal of a pocketed ball, setting a stopped velocity to zero) are continuous where the relevant clock is the unique active one. The evolution up to any fixed number of events is therefore a finite composition of continuous maps, and it is locally the same composition: sufficiently small perturbations of x^* produce the same finite sequence of event types involving the same balls. In particular the set of pocketed balls, and so $N(c, \cdot)$, is locally constant at x^*. □

The lemma isolates all the analysis; what remains is engineering.

Lemma 2 (Witness)**.** *There exist an admissible configuration c^* and a shot $x^* \in \Omega$, regular for c^*, with $N(c^*, x^*) = 21$.*

Construction. The witness is a *single-carrier chain*. The cue ball is the only carrier: it strikes twenty-one object balls in succession, and at each collision the struck ball departs along the line of centers aimed at a pocket center while the cue is deflected onward to the next ball. No object ball ever strikes another; every pocketing is a struck ball sent directly to a pocket by the cue, and the cue itself is not pocketed. That accounts for all twenty-one object balls in twenty-one collisions.

Two facts make the chain reach length twenty-one. First, at a cut through angle θ (the angle between the cue's incoming velocity and the line of centers) the struck ball leaves with speed factor $\frac{1+e}{2}\cos\theta$ and the cue retains the complementary tangential component; a *thick* cut (θ near $\pi/2$) therefore costs the cue almost none of its speed while still sending the struck ball off with enough to reach a pocket. Choosing thick cuts lets a single cue survive all twenty-one collisions, using cushion rebounds between strikes to reach balls in every region of the table. Second, each ball is placed *after* its predecessors are fixed, by reading the cue's actual post-collision state from an event-driven integrator of Definition 2 and solving, by elementary root finding, for the ball position whose line of centers points at a chosen pocket. Because every placement is chosen against, and then re-verified in, the same integrator that defines the dynamics, the planned collision sequence and all twenty-one pocketings are exact by construction rather than by an independent analytic prediction that might drift from the dynamics. All margins (clearances between balls not meant to touch, gaps between event times, entry speeds into pockets) are strictly positive, and the shot x^* is regular in the sense above. Appendix A lists the coordinates, the shot, the pocketing log, and the verified margins. □

Theorem 1. *There is an admissible configuration c^* for which $\{ x \in \Omega : N(c^*, x) = 21 \}$ contains a nonempty open set, and hence has positive Lebesgue measure.*

Proof. Immediate from Lemmas 1 and 2. □

Remark 3 (Status of the verification). The witness is verified numerically, in double precision: the shot x^* is replayed in an event-driven integrator of Definition 2 and produces exactly twenty-one pocketings, with the individual margins that enter regularity (pocket-entry speeds, inter-ball clearances, event-time gaps) all strictly positive and many orders of magnitude above the rounding scale. Regularity of x^* is what Lemma 1 consumes, and it yields an open neighborhood of x^* on which $N \equiv 21$; that neighborhood is what carries the positive-measure conclusion. A purist can upgrade the verification to a rigorous certificate by interval arithmetic: the trajectory is a finite composition of explicit algebraic maps, so enclosing it is routine, and the margins leave the intervals ample room. We report the double-precision margins and flag the theorem as computer-assisted.

Remark 4 (Positive measure versus observable width). It is worth stating plainly what the theorem does and does not quantify. Lemma 1 guarantees that the success set around a regular witness is open, hence of positive Lebesgue measure; it says nothing about how large that measure is. For our twenty-one-collision witness the guaranteed neighborhood is in fact extraordinarily thin: scanning the shot parameters (V_0, φ) outward from x^*, the interval on which all twenty-one balls still fall shrinks below the resolution of a double-precision grid (a change of order 10^{-6} m/s in V_0 already spoils the clearance). This is not a defect of the proof but a feature of long collision chains: sensitivity compounds multiplicatively across twenty-one successive cuts, so the regular set, while genuinely open, is unobservably small. Shorter chains built by the same construction have visibly fat success sets—a three-ball chain, for instance, clears over a V_0-window of order 10^{-1} m/s, five orders of magnitude wider—which is exactly the qualitative gap between "measure zero" and "unobservably small" that the introduction promised, now made concrete at both ends. The positive-measure statement of Theorem 1 is therefore mathematically robust and physically fragile, and both halves matter: the total clearance is possible in principle, and invisible in practice.

4 The opening break

Conjecture 1. *$\{ x \in \Omega : N(c_0^{\varepsilon}, x) = 21 \}$ is nonempty (and then, by Lemma 1 applied at a regular witness, of positive measure) for $\varepsilon = 0.5$ mm.*

Why should this be true? Nothing in the construction of Lemma 2 used freedom to place balls generously; it used freedom to place them exactly. The opening configuration is a single point in configuration space, but the shot space still offers a two-parameter family, and the dynamics from a tight pack is ferociously rich: the first impact sprays fifteen reds into a recirculating cascade of secondary collisions. Richness is not a proof, and we are honest that the conjecture is open even at the level of overwhelming numerical evidence: no simulated stroke below comes remotely close to twenty-one.

Two computational points deserve separation. First, the conjecture is *decidable in principle by simulation*: a single stroke x^* found by any search whatsoever, then certified by interval arithmetic to be regular with $N = 21$, proves it, by Lemma 1. The obstacle is not verification but search: success sets, if nonempty, live at the bottom of a chaotic landscape in which each additional required pocketing multiplies the target measure by something like the factors estimated in Section 5, and twenty-plus collisions of sensitivity compound beyond what double precision can even represent reliably along the way. Second, the conjecture is *not decidable by direct Monte Carlo*: no feasible number of random strokes distinguishes an event of probability 10^{-40} from an impossible one. Sampling can only ever bound $P(21)$ from above; a single certified trajectory settles existence. The asymmetry between the two computations is the heart of the matter.

5 How rare is rare?

We estimate $P(k)$, the probability that a stroke drawn uniformly from Ω pockets exactly k object balls from c_0^{ε}, by direct simulation of Model M with an event-driven integrator (the same code that verifies the witness; this is deliberate, so that every number in the paper refers to one and the same dynamical system). Over $n = 30{,}000$ independent strokes with a fixed seed, the counts fall off cleanly and monotonically; intervals below are Wilson score intervals at 95%.

k	count	$\hat{P}(k)$	95% Wilson CI
0	18489	6.16e-01	[6.11e-01, 6.22e-01]
1	7811	2.60e-01	[2.55e-01, 2.65e-01]
2	2564	8.55e-02	[8.24e-02, 8.87e-02]
3	814	2.71e-02	[2.54e-02, 2.90e-02]
4	251	8.37e-03	[7.40e-03, 9.46e-03]
5	58	1.93e-03	[1.50e-03, 2.50e-03]
6	10	3.33e-04	[1.81e-04, 6.14e-04]
7	3	1.00e-04	[3.40e-05, 2.94e-04]

Three comments on reading the numbers. First, the log-linear decay visible over the populated range (a straight line of slope $b \approx -0.57$ per ball on the $\log_{10}$ scale, the largest observed value being $k = 7$) invites extrapolation to $k = 21$, and we report the extrapolated value, $\sim 10^{-12}$, but its logical status must be kept straight: extrapolation cannot distinguish a tiny positive $P(21)$ from an exact zero, so the number is meaningful only conditional on Conjecture 1, and even then it assumes the decay law persists through a regime the data never probes. Second, the estimates depend on model parameters that are only physically approximate; a sensitivity grid over the friction rate $a \in \{0.05, 0.10, 0.15\}$ and the restitution $e \in \{0.90, 0.95, 0.98\}$

confirms that the qualitative decay, which is all the argument uses, is stable: every one of the nine cells shows a steep negative log-linear slope with an excellent fit.

a	e	slope b	R^2	max k
0.05	0.9	-0.427	0.976	6
0.05	0.95	-0.356	0.991	7
0.05	0.98	-0.333	0.976	7
0.1	0.9	-0.554	0.992	7
0.1	0.95	-0.504	0.985	5
0.1	0.98	-0.458	0.986	6
0.15	0.9	-0.642	0.997	5
0.15	0.95	-0.686	0.999	5
0.15	0.98	-0.623	0.989	5

Higher friction steepens the decay, as one would expect (less energy in the system reaches fewer balls), but in no cell does the exponential character weaken. Third, the largest k observed in the run is itself a useful falsifiable benchmark: we invite readers to beat the record shot (seed and parameters in the package) with any search method; every improvement is a data point on the road to Conjecture 1.

Direct estimation of $P(21)$ is not merely hard but structurally hopeless: at any rate suggested by the observed decay, the expected number of successes over centuries of continuous simulation is far below one. The honest route to smaller probabilities is rare-event methodology, for instance subset simulation through the nested levels $\{N \geq k\}$, estimating conditional factors $P(N \geq k+1 \mid N \geq k)$ one at a time. We propose this, and the search for a certified break witness, as the two serious follow-up computations.

6 Discussion

The folklore is best restated as three true sentences. A total clearance in one stroke is possible in the model, and forms a set of positive measure there—robust in principle, though for the twenty-one-ball witness that robustness is of unobservably small width (Theorem 1 and the remark following it). From the opening break it is, we believe, also possible, and settling that belief is a well-posed finite computation that current search methods cannot yet perform (Section 4). And it will never be observed, because its probability, if positive, is protected by an exponential decay that our experiments measure in its shallow end (Section 5). The bar bet, in other words, confuses measure zero with unobservably small. Trick-shot artists have always lived in the gap between the two: an engineered multi-ball finish is nothing but a witness configuration with humanly achievable margins, and the single-carrier chain of Lemma 2 is a formalized trick shot—one whose margins are real but, at twenty-one balls, far too fine for any human cue. What mathematics adds is the guarantee that the margins exist at all, and a clear account of why the guarantee for the break position, claim (ii), remains open: it sits exactly where verification is easy and search is hopeless, a combination that will be familiar to anyone who has thought about needles in high-dimensional haystacks.

A The witness

Model instance. $L = 3.569$, $W = 1.778$, $R = 0.02625$, $a = 0.10$, $e = 0.95$, $e_c = 0.90$, $\rho = 0.055$ (SI).

The shot. Cue at (0.5000, 0.7318), $V_0 = 11.9$ m/s, $\varphi = 0.00442$ rad.

#	x (mm)	y (mm)	pocket
1	1666.0	789.0	M-t
2	3271.5	601.3	C-tr
3	3519.6	434.6	C-br
4	1941.1	622.8	M-b
5	400.8	1005.1	C-bl
6	51.6	1249.7	C-tl
7	1626.3	1096.5	M-t
8	1866.9	934.3	M-b
9	3267.0	909.8	C-tr
10	3513.9	719.2	C-br
11	2027.5	833.1	M-b
12	567.2	1251.2	C-bl
13	369.3	1665.7	M-b
14	1202.0	1203.7	M-t
15	1974.8	278.1	M-b
16	2175.5	266.3	C-tr
17	2983.8	473.5	C-br
18	3377.6	1550.5	C-tr
19	3175.6	1590.0	C-bl
20	3110.9	1729.6	C-tl
21	3178.0	1422.8	C-br

Margins. All 21 object balls pocketed (verified end to end); minimum pairwise ball separation 154 mm; pocket-entry speeds from 0.14 to 4.27 m/s.

The verification protocol: the layout is replayed in the event-driven integrator; we require that exactly the planned collision pairs occur, that all twenty-one object balls are pocketed, and we report the minimum gap between consecutive events, the minimum pocket-entry speed, and the minimum clearance attained by any pair of balls not designed to touch. All three margins are far above double-precision rounding scales, which is the sense in which Theorem 1 is computer-assisted; an interval-arithmetic certificate would be a finite, routine upgrade.

B Reproduction

All results are produced by short, dependency-free Python scripts around a single event-driven implementation of Definition 2: `model_sim.py` (the engine, cross-validated against an independent reference implementation), `witness_bv.py` (the build-and-verify constructor, which grows the chain one collision at a time inside the engine and emits the witness of Appendix A), `mc_break.py` (the Monte Carlo, the sensitivity grid, and the table above), and `make_figures.py`. Seeds are fixed in the scripts. Readers who want a full-physics cross-check including spin and sliding friction can replay the witness in an engine such as pooltool [4]; by Remark 1 the zero-spin witness is the right object to test.

References

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